Exact symplectic comparison under weak taming

A mathematical proof blueprint

Exact Liouville comparison project

24 October 2026

Abstract

We give the four-path Moser proof of the radial exact comparison theorem. The argument constructs a small radial coefficient from the inverse-form growth bound and a large coefficient dominating an otherwise unrestricted primitive. A scalar exhaustion controls the first and third flows; two different barriers control the forward and reverse directions of the interpolation flows. We then prove the exact pullback identity with its smooth integral gauge and record the composition order explicitly. We also construct a smooth cylindrical compatible structure and extend arbitrary prescribed compatible contact data, preserving the entire closed end. The final sections derive the finite-type Liouville, properly embedded Stein, and common-reference applications. All estimates use the stated reference metric.

1 The theorem and the architecture of its proof

Let X be a Hausdorff second-countable smooth real manifold without boundary, of one fixed arbitrary dimension $d \geq 0$. Fix a smooth one-form β with $\Omega = \mathrm{d}\beta$ symplectic and an Ω-compatible smooth almost complex structure J. Put $g = \Omega(\,\cdot\,, J\,\cdot\,)$. Assume that $R : X \to [1, \infty)$ is smooth and proper, $c > 0$, $r_* > 1$, and, on $\{R > r_*\}$,

$$\beta = -c\,\mathrm{d}R \circ J, \qquad a = c|\mathrm{d}R|^2_{g^*} \leq R. \tag{1}$$

The compact core $\{R \leq r_*\}$ is exempt from these radial identities.

Let λ be smooth, with $\omega = \mathrm{d}\lambda$ symplectic and

$$\omega(v, Jv) \geq 0 \qquad (v \in TX).$$

Assume that for some smooth $B : [1, \infty) \to (0, \infty)$ and $C > 0$,

$$B(r) \geq r, \qquad \int_1^\infty \frac{r}{B(r)^2}\,\mathrm{d}r = +\infty, \qquad \|\omega^{-1}\|_g \leq CB(R) \tag{2}$$

outside a compact set, with the first two conditions holding on the whole half-line. The inverse norm is the operator norm of $(v \mapsto \omega(v, \cdot))^{-1}$ from the g-dual to the g-tangent space.

Theorem 1.1. *There exist a diffeomorphism $F : X \to X$, smoothly isotopic to the identity, and a globally smooth $h : X \to \mathbb{R}$ such that*

$$F^*\lambda = \beta + \mathrm{d}h, \qquad F^*\omega = \Omega.$$

No bound on the primitive λ is assumed. Neither completeness of g nor completeness of either Liouville field is used.

We construct positive radial functions k and A, constant on $[1, r_*]$, and compare the five primitives

$$\lambda \;\longrightarrow\; \lambda + k(R)\beta \;\longrightarrow\; \lambda + A(R)\beta \;\longrightarrow\; A(R)\beta \;\longrightarrow\; \beta. \tag{3}$$

The actual forward paths used to construct the maps are

$$\begin{aligned} \mu_t^{(1)} &= \lambda + t\,k(R)\beta, & \mu_t^{(2)} &= \lambda + \big((1-t)k(R) + tA(R)\big)\beta, \\ \mu_t^{(3)} &= A(R)\beta + t\lambda, & \mu_t^{(4)} &= \big(1 + t(A(R)-1)\big)\beta. \end{aligned} \tag{4}$$

The third and fourth arrows in (3) use the inverses of the corresponding forward endpoint maps. We retain this orientation throughout the exactness calculation.

2 Two pointwise geometric estimates

2.1 Weak taming and the resolvent contraction

Lemma 2.1. *Let η be a J-compatible symplectic form on a finite-dimensional real vector space, and put $h = \eta(\,\cdot\,, J\,\cdot\,)$. If a nondegenerate alternating form ω satisfies $\omega(v, Jv) \geq 0$, then $\omega + t\eta$ is nondegenerate for every $t \geq 0$. Moreover,*

$$\iota_X(\omega + t\eta) = \iota_W\omega \quad\Longrightarrow\quad |X|_h \leq |W|_h.$$

Proof. Define C by $\omega(v, w) = h(Cv, w)$ and put $T = -JC$. Compatibility gives $\eta(v, w) = h(Jv, w)$, $J^* = -J$, and

$$h(Tv, v) = h(Cv, Jv) = \omega(v, Jv) \geq 0.$$

Consequently

$$|(T + t\mathrm{id})v|_h^2 = |Tv|_h^2 + 2t\,h(Tv, v) + t^2|v|_h^2 \geq |Tv|_h^2 + t^2|v|_h^2.$$

For $t > 0$, this proves injectivity of $T + t\mathrm{id}$ and hence invertibility by finite dimensionality. At $t = 0$, invertibility follows from nondegeneracy of ω. Since $C + tJ = J(T + t\mathrm{id})$, this proves the first assertion.

The displayed inequality also gives $\|T(T + t\mathrm{id})^{-1}\|_h \leq 1$. The operators T and $(T + t\mathrm{id})^{-1}$ commute, because T commutes with $T + t\mathrm{id}$. The contraction equation is $(T + t\mathrm{id})X = TW$, and therefore

$$|X|_h = |(T + t\mathrm{id})^{-1}TW|_h \leq |W|_h.$$

The argument uses no strictly positive lower bound for $\omega(v, Jv)$. □

2.2 The metric of a radial primitive

Let $\rho : [1, \infty) \to (0, \infty)$ be smooth and constant on $[1, r_*]$. Write

$$\eta_\rho = \mathrm{d}(\rho(R)\beta), \qquad h_\rho(v, w) = \eta_\rho(v, Jw).$$

On the core, $h_\rho = \rho(R)g$, because the coefficient derivative vanishes there. Outside the core, the product rule and (1) give

$$h_\rho(v, w) = \rho g(v, w) + c\rho'\big(\mathrm{d}R(v)\mathrm{d}R(w) + \mathrm{d}R(Jv)\mathrm{d}R(Jw)\big), \tag{5}$$

where coefficients on X are evaluated at R.

Lemma 2.2. *If $\rho(r) + r\rho'(r) > 0$ for all $r \geq 1$, then η_ρ is J-compatible. On the exterior, put $D_\rho = \rho + \rho' a$. Then*

$$D_\rho > 0, \qquad |\mathrm{d}R|^2_{h_\rho^*} = \frac{a}{cD_\rho}, \qquad |\beta|^2_{g^*} = ca. \tag{6}$$

If $\rho' \leq 0$, then $(\rho + R\rho')g \leq h_\rho \leq \rho g$. If $\rho' \geq 0$, then $h_\rho \geq \rho g$. These metric inequalities hold on the whole manifold.

Proof. The right side of (5) is symmetric and invariant under simultaneous application of J. At a point with $\mathrm{d}R \neq 0$, decompose the tangent space into the g-orthogonal plane spanned by $\nabla_g R, J\nabla_g R$ and its orthogonal complement. The eigenvalues of h_ρ relative to g are D_ρ on that plane and ρ on the complement. If $\rho' \leq 0$, then $a \leq R$ gives $D_\rho \geq \rho + R\rho' > 0$; if $\rho' \geq 0$, then $D_\rho \geq \rho > 0$. At $\mathrm{d}R = 0$ the added tensor vanishes and $h_\rho = \rho g$. These statements prove positivity and the metric inequalities. The exterior derivative η_ρ is closed, so positivity and J-invariance prove compatibility and symplecticity.

The covector $\mathrm{d}R$ is supported on the distinguished plane; its squared dual norm is thus divided by D_ρ. Since $|\mathrm{d}R|^2_{g^*} = a/c$, this gives the first norm identity. The second follows from $\beta = -c\,\mathrm{d}R \circ J$ and the g-orthogonality of J. Both identities hold at $\mathrm{d}R = 0$ as well. On the core, $\rho' = 0$, which proves the remaining global metric claims without using the exterior radial identities there. □

3 Constructing the two radial coefficients

3.1 A small coefficient adapted to the inverse-form bound

Fix a smooth nondecreasing step function $\vartheta : \mathbb{R} \to [0,1]$ which is zero on $(-\infty, 0]$ and one on $[1, \infty)$. For example, with $q(t) = e^{-1/t}$ for $t > 0$ and $q(t) = 0$ otherwise, take $\vartheta(t) = q(t)/(q(t) + q(1-t))$. Define

$$S(r) = 1 + \int_1^r \frac{u}{B(u)^2}\,\mathrm{d}u, \qquad \chi(r) = \vartheta(r - r_*).$$

Then $S \geq 1$, $S' = r/B(r)^2 > 0$, $S(r) \to \infty$, and $S(r) \leq 1 + \log r$, using $B(r) \geq r$. Set

$$k(r) = \exp\left(\int_1^r \chi(s)\left(\frac{S'(s)}{2S(s)} - \frac{1}{s}\right)\mathrm{d}s\right), \qquad m(r) = k(r) + rk'(r). \tag{7}$$

Since $rS'(r)/(2S(r)) \leq 1/2$, direct differentiation gives

$$\begin{aligned} &0 < k \leq 1, \qquad k = 1 \text{ on } [1, r_*], \qquad k' \leq 0, \\ &m = k\left(1 - \chi + \frac{\chi r^2}{2B(r)^2 S(r)}\right) > 0. \end{aligned} \tag{8}$$

Strict positivity of the last expression holds both when $\chi < 1$ and when $\chi = 1$.

For $r \geq r_* + 1$, the integrand in (7) is the derivative of $\frac{1}{2}\log S - \log r$. Thus there is $b_0 > 0$ such that

$$k(r) = b_0 \frac{\sqrt{S(r)}}{r}, \qquad m(r) = \frac{b_0 r}{2B(r)^2\sqrt{S(r)}}. \tag{9}$$

In particular $rk(r) \to \infty$. Lemma 2.2 now supplies a compatible symplectic form $\eta = \mathrm{d}(k(R)\beta)$ and metric $h = \eta(\,\cdot\,, J\,\cdot\,)$ satisfying

$$m(R)g \leq h \leq k(R)g. \tag{10}$$

This construction works for the given arbitrary admissible B.

3.2 A large coefficient with no upper growth restriction

The continuity of the g-dual norm of λ and properness of R give finite nondecreasing numbers

$$M_n = 1 + \sup_{\{R \le r_* + n + 1\}} |\lambda|_{g^*} \qquad (n = 0, 1, 2, \ldots).$$

For an empty sublevel set, the displayed supremum is defined as zero. Define

$$A(r) = M_0 + \sum_{j=1}^{\infty} (M_j - M_{j-1})\, \vartheta(r - r_* - j + 1). \tag{11}$$

On every bounded interval, all sufficiently large-index summands vanish. The sum is therefore locally finite and smooth. Every coefficient is nonnegative, so $A' \ge 0$; also $A = M_0$ on $[1, r_*]$.

If $r \in [r_* + n, r_* + n + 1]$, each summand with $1 \le j \le n$ has reached its full value, hence $A(r) \ge M_n$. The sublevel defining M_n contains every point with this value of R. For $R \le r_*$, M_0 already controls the primitive. Consequently

$$A \ge 1 \ge k, \qquad A + rA' > 0, \qquad A(R(x)) \ge 1 + |\lambda_x|_{g^*}. \tag{12}$$

There is no upper bound on the growth of A. The interpolation argument below is designed to allow this freedom.

4 Global time-dependent flows from proper barriers

For a smooth path of primitives μ_t whose differentials are symplectic, the Moser field is uniquely defined by

$$\iota_{X_t} \mathrm{d}\mu_t = -\dot{\mu}_t. \tag{13}$$

In local frames it is the inverse of a smooth invertible matrix applied to a smooth covector. Hence it is jointly smooth on the open nondegeneracy locus. When nondegeneracy holds on $[0, 1] \times X$, this locus supplies a neighborhood of the whole closed strip; no uniform extra time interval on the noncompact manifold is required.

Lemma 4.1 (Compact confinement and continuation)**.** *Let X_t be such a smooth time-dependent field. If every solution through a fixed initial time and point remains in a fixed compact subset of X as its time varies over its interval of existence inside $[0, 1]$, then that solution exists on the whole interval, in both directions from the initial time. If the confinement hypothesis holds for every initial datum, the solution maps $E_{s,t}$ are smooth diffeomorphisms and satisfy*

$$E_{s,s} = \mathrm{id}, \qquad E_{s,t} \circ E_{r,s} = E_{r,t}, \qquad E_{s,t}^{-1} = E_{t,s}.$$

They depend jointly smoothly on s, t, x up to the endpoints.

Proof. Local existence, uniqueness, and smooth dependence follow in charts from the smooth Euclidean ODE theorem. Around the compact spacetime set $[0, 1] \times K$ containing a confined curve, finitely many local existence neighborhoods give a common positive lifetime for nearby initial data. If a maximal solution had a finite endpoint within the prescribed time interval, restarting it sufficiently close to that endpoint would extend it past the endpoint. Uniqueness identifies the new solution with the old one on their overlap, a contradiction. The argument is identical in reverse time.

Uniqueness gives the evolution identities and inverse maps. Along any fixed finite trajectory, finitely many local flow maps compose to give smooth dependence on both times and the

initial point. Equivalently, use the local flow of the spacetime field $\partial_t + X_t$ on its open domain. These local descriptions agree by uniqueness and give the asserted smooth evolution, including one-sided endpoint derivatives and local smooth extensions around the closed time square. □

Corollary 4.2 (One proper scalar)**.** *If a smooth proper function $P : X \to \mathbb{R}$ is bounded below and $|X_t P| \le D$ on $[0,1] \times X$, then the conclusions of Lemma 4.1 hold. An estimate valid outside one fixed compact set is sufficient.*

Proof. Along a trajectory starting at (s,x), $|P(\gamma(t)) - P(x)| \le D|t-s|$. Its image lies in the compact inverse image of a bounded closed interval. If the estimate is initially exterior, continuity of $(t,x) \mapsto X_t P(x)$ on the product of the compact time interval and the exceptional compact set gives one enlarged global bound. □

This uses properness, not metric completeness. In particular, a bound on the norm of a field is used below only after conversion to a bound on the derivative of a proper scalar.

5 The interpolation lemma and its reverse-time barrier

Let σ be a smooth one-form with $\mathrm{d}\sigma(v, Jv) \ge 0$; its differential may be degenerate. Let ρ_0, ρ_1 be positive smooth functions, constant on $[1, r_*]$, with

$$\rho_i + r\rho_i' > 0, \qquad \rho_1 \ge \rho_0, \qquad r\rho_0(r) \longrightarrow \infty. \tag{14}$$

Put $\delta = \rho_1 - \rho_0 \ge 0$ and $\rho_t = (1-t)\rho_0 + t\rho_1$.

Lemma 5.1 (Radial interpolation)**.** *The path $\mu_t = \sigma + \rho_t(R)\beta$ has symplectic differential. Its Moser evolution exists for all pairs of times in $[0,1]$, with the smoothness, inverse, and evolution properties of Lemma 4.1.*

Proof. The coefficient inequalities persist under convex combinations. Define the compatible metric

$$h_t = \mathrm{d}(\rho_t(R)\beta)(\,\cdot\,, J\,\cdot\,).$$

Lemma 2.2 and weak taming give

$$\mathrm{d}\mu_t(v, Jv) = \mathrm{d}\sigma(v, Jv) + |v|^2_{h_t} > 0 \quad (v \ne 0),$$

which proves nondegeneracy. Closedness is automatic. The Moser equation is $\iota_{X_t}\mathrm{d}\mu_t = -\delta(R)\beta$.

On the exterior, evaluate it on JX_t:

$$|X_t|^2_{h_t} + \mathrm{d}\sigma(X_t, JX_t) = -c\delta\, X_t R. \tag{15}$$

If $\delta = 0$, nondegeneracy implies $X_t = 0$ at the point. Otherwise (15) gives $Y := -X_t R \ge 0$. Using (6), with $D_t = \rho_t + \rho_t' a$, yields

$$Y^2 \le |\mathrm{d}R|^2_{h_t^*} |X_t|^2_{h_t} \le \frac{a}{cD_t}\, c\delta Y.$$

For $Y > 0$ divide by Y; for $Y = 0$ the result is immediate. Thus

$$0 \le -X_t R \le \frac{\delta a}{D_t} \quad \text{on } \{R > r_*\}. \tag{16}$$

Define the moving scalar $Q(t,x) = R(x)\rho_t(R(x))$. Its total derivative along the field satisfies, on the exterior,

$$\begin{aligned}(\partial_t + X_t)Q &= \delta R + (\rho_t + R\rho_t')X_tR \\ &\geq \delta\left(R - \frac{a(\rho_t + R\rho_t')}{D_t}\right) = \frac{\delta\rho_t(R-a)}{D_t} \geq 0. \end{aligned} \tag{17}$$

The sign of $\rho_t + R\rho_t'$ is positive, so using the lower bound for X_tR in this calculation is legitimate.

Compactness of the core and the time interval supplies constants $C_0, C_1 \geq 0$ such that, globally,

$$X_tR \leq C_0, \qquad (\partial_t + X_t)Q \geq -C_1.$$

For a trajectory with $\gamma(s) = x$ and $t \geq s$, $R(\gamma(t)) \leq R(x) + C_0(t-s)$. For $t \leq s$, integrate the other inequality to obtain

$$Q(t,\gamma(t)) \leq Q(s,x) + C_1(s-t).$$

But $Q(t,y) \geq R(y)\rho_0(R(y))$, and the latter expression tends to infinity with $R(y)$ by (14). This also bounds $R(\gamma(t))$, now in reverse time. Both portions of the trajectory lie in a single compact R-sublevel, so Lemma 4.1 applies. □

Remark 5.2 (Why the two barriers matter)**.** The argument does not assert a bound on $|X_tR|$ independent of R. The forward bound controls R, while the reverse bound controls Q. Arbitrarily fast growth of the upper coefficient ρ_1 does not spoil the lower estimate $Q(t,x) \geq R(x)\rho_0(R(x))$. If one constructs trajectories using compactly supported cutoffs, these estimates remain available: for $0 \leq \zeta \leq 1$,

$$(\partial_t + \zeta X_t)Q = \zeta(\partial_t + X_t)Q + (1-\zeta)\partial_t Q,$$

and $\partial_t Q = \delta(R)R \geq 0$. Thus the cutoff does not destroy the reverse-time inequality.

6 The first path: the integral growth hypothesis

Consider $\mu_t = \lambda + t\,k(R)\beta$. Lemma 2.1, applied to $\eta = \mathrm{d}(k(R)\beta)$, proves that $\mathrm{d}\mu_t = \omega + t\eta$ is symplectic. Define W by $\iota_W\omega = -k(R)\beta$. The actual Moser field satisfies

$$\iota_{X_t}(\omega + t\eta) = \iota_W\omega, \qquad |X_t|_h \leq |W|_h.$$

On the region where the inverse-form bound holds,

$$|W|_g \leq CB(R)k(R)|\beta|_{g^*}.$$

The metric comparison (10) then gives

$$|X_t|_g \leq \sqrt{\frac{k(R)}{m(R)}}\,|W|_g.$$

On the radial exterior, $|\mathrm{d}R|_{g^*}|\beta|_{g^*} = a \leq R$ by (6). For $R \geq r_* + 1$, outside the original exceptional compact set,

$$\begin{aligned}|X_tR| &\leq CB(R)k(R)R\sqrt{\frac{k(R)}{m(R)}} \\ &= \sqrt{2}\,Cb_0\,\frac{B(R)^2S(R)}{R}. \end{aligned} \tag{18}$$

Here the tail formulas (9) were used exactly.

Now set $P = \log S(R)$. Since $S \geq 1$ and $S(r) \to \infty$, this is a smooth proper function bounded below by zero. For example, a bounded upper value of $\log S(R)$ bounds R, and closed sublevels are compact by properness of R. Using $S' = r/B(r)^2$ in (18) gives

$$|X_t P| \leq \sqrt{2}\, C b_0.$$

The region excluded from this estimate is contained in the union of the original compact exception and the compact set $\{R \leq r_* + 1\}$. Smoothness supplies a bound there too. Corollary 4.2 therefore constructs the full two-time smooth evolution of the first path.

This is the only stage whose completeness estimate uses the inverse-form growth bound. Its right-hand side is $-k(R)\beta$, so no size bound on λ enters the argument.

7 The other three paths

7.1 The second path

Apply Lemma 5.1 with

$$\sigma = \lambda, \qquad \rho_0 = k, \qquad \rho_1 = A.$$

Weak taming gives the hypothesis on $\mathrm{d}\sigma$. Equations (8), (9), and (12) give all coefficient hypotheses, including $rk(r) \to \infty$. This constructs the smooth evolution for $\mu^{(2)}$ in (4). Its reverse-time barrier is bounded below by $Rk(R)$ independently of how quickly A grows.

7.2 The third path

Use the forward path $\mu_s = A(R)\beta + s\lambda$. Since $A' \geq 0$, Lemma 2.2 gives

$$\mathrm{d}\mu_s(v, Jv) = \mathrm{d}(A(R)\beta)(v, Jv) + s\omega(v, Jv) \geq A(R)|v|_g^2.$$

Thus every differential is symplectic, including at $s = 0$. The Moser equation is $\iota_{X_s}\mathrm{d}\mu_s = -\lambda$. At JX_s it implies

$$A(R)|X_s|_g^2 \leq -\lambda(JX_s) \leq |\lambda|_{g^*}|X_s|_g.$$

If $X_s = 0$, the following bound is immediate; otherwise divide by its positive norm. Equation (12) yields

$$|X_s|_g \leq \frac{|\lambda|_{g^*}}{A(R)} \leq 1. \tag{19}$$

Use the proper smooth scalar $P = \sqrt{R}$, rather than a metric completeness assertion. On the radial exterior,

$$|X_s P| \leq \frac{|\mathrm{d}R|_{g^*}}{2\sqrt{R}}|X_s|_g \leq \frac{1}{2\sqrt{c}}.$$

The derivative is bounded on the core by joint continuity and compactness. Corollary 4.2 constructs the full evolution in both directions. The forward endpoint goes from the primitive $A(R)\beta$ to $A(R)\beta + \lambda$ in the Moser pullback sense. Its inverse supplies the third arrow of (3).

7.3 The fourth path

Apply Lemma 5.1 with $\sigma = 0$, $\rho_0 = 1$, and $\rho_1 = A$. The lower divergence condition is simply $r \to \infty$. It constructs the evolution for $\mu_t^{(4)} = (1 + t(A(R) - 1))\beta$, from β to $A(R)\beta$. The inverse endpoint supplies the fourth arrow.

For each of the four paths let $G_{j,t} = E_{0,t}^{(j)}$, and write $G_j = G_{j,1}$. Every $G_{j,t}$ and its inverse depend smoothly on (t, x), with $G_{j,0} = \mathrm{id}$. All four endpoint maps are therefore smoothly isotopic to the identity, as are their inverses.

8 The exact pullback identity and its smooth gauge

We prove the identity for any of the smooth Moser paths above. Write μ_t, X_t, and G_t for its primitive, field, and identity-start evolution.

Lemma 8.1 (Time-dependent pullback formula)**.** *For $0 \le t \le 1$,*

$$\frac{\mathrm{d}}{\mathrm{d}t} G_t^* \mu_t = G_t^*(\dot{\mu}_t + \iota_{X_t} \mathrm{d}\mu_t) + \mathrm{d}\big(\mu_t(X_t) \circ G_t\big).$$

Derivatives at the two endpoints are interpreted within $[0, 1]$.

Proof. Fix x and $v \in T_x X$. In local coordinates along the trajectory, put $y = G_t(x)$ and $w = \mathrm{d}G_t(x)v$. Differentiating $\mu_t(y)(w)$ gives three terms:

$$\dot{\mu}_t(y)(w) + (D\mu_t(y)[X_t(y)])(w) + \mu_t(y)(DX_t(y)[w]).$$

The equation for the differentiated trajectory, $\partial_t w = DX_t(y)[w]$, follows by differentiating $\partial_t G_t = X_t \circ G_t$ in the initial point and commuting mixed derivatives of the smooth local expression for G. For a one-form in coordinates,

$$\mathrm{d}\mu_t(X_t, w) = (D\mu_t[X_t])(w) - (D\mu_t[w])(X_t).$$

Meanwhile the derivative in direction v of $\mu_t(X_t) \circ G_t$ is $(D\mu_t[w])(X_t) + \mu_t(DX_t[w])$. Combining these formulas proves the identity. Each expression is intrinsic, so the local computations agree on chart overlaps. Smoothness up to the endpoints supplies the corresponding one-sided identities there. □

Substituting the actual Moser equation (13) cancels the first two terms:

$$\frac{\mathrm{d}}{\mathrm{d}t} G_t^* \mu_t = \mathrm{d}\big(\mu_t(X_t) \circ G_t\big). \tag{20}$$

Lemma 8.2 (Smooth parameter integration)**.** *If $f : [0, 1] \times X \to \mathbb{R}$ is jointly smooth up to the endpoints, then $H(x) = \int_0^1 f(t, x)\,\mathrm{d}t$ is smooth, and*

$$\mathrm{d}H_x(v) = \int_0^1 \mathrm{d}_x f(t, x)(v)\,\mathrm{d}t.$$

Proof. Work in a coordinate chart about a fixed point x_0. Smoothness up to the endpoints gives local smooth extensions. Compactness of $[0, 1] \times \{x_0\}$ permits finitely many such neighborhoods and then a smaller spatial neighborhood on which the relevant derivatives are defined throughout the time interval. Choose its closure inside a relatively compact coordinate neighborhood. Each fixed finite-order spatial derivative is continuous and hence bounded on the resulting compact product.

For the first derivative, apply the fundamental theorem of calculus to the spatial difference quotient. Continuity of the derivative on that compact product gives uniform convergence as the increment tends to zero, allowing integration in t to pass to the limit. This gives the displayed derivative formula. Repeating the same argument for every finite order proves C^∞ regularity. Coordinate changes obey the usual chain rule, so the local smooth functions and their differentials give the stated global result. □

Only local bounds near each spatial point are needed. There is no global bound on derivatives of f over the noncompact manifold and no assumption of a uniform extra time strip. Assigning arbitrary values to f outside $[0,1]$ when writing the integral does not assert smoothness of that arbitrary extension.

Proposition 8.3 (Exactness of one endpoint)**.** *Define the actual scalar function*

$$H(x) = \int_0^1 \mu_t(X_t)(G_t(x))\,\mathrm{d}t.$$

Then H is smooth and

$$G_1^*\mu_1 = \mu_0 + \mathrm{d}H, \qquad G_1^*\mathrm{d}\mu_1 = \mathrm{d}\mu_0. \tag{21}$$

Proof. The integrand is jointly smooth by smoothness of the form, the vector field, and the evolution. Lemma 8.2 proves smoothness of H and the derivative-interchange formula. For fixed x, v, integrate (20) from zero to one and use $G_0 = \mathrm{id}$. The fundamental theorem of calculus then gives the primitive equality evaluated on v. Taking exterior derivatives, using naturality of d under pullback and $\mathrm{d}^2 H = 0$, gives the two-form equality. □

Thus the gauge is constructed by the displayed integral; its existence or smoothness is not an additional premise.

9 Composition, inversion, and the final isotopy

For clarity, write the five primitives in (3) as

$$\nu_0 = \lambda, \quad \nu_1 = \lambda + k(R)\beta, \quad \nu_2 = \lambda + A(R)\beta, \quad \nu_3 = A(R)\beta, \quad \nu_4 = \beta.$$

Applying Proposition 8.3 to the four forward paths gives smooth H_1, H_2, H_3, H_4 with

$$\begin{aligned} G_1^*\nu_1 &= \nu_0 + \mathrm{d}H_1, \quad G_2^*\nu_2 = \nu_1 + \mathrm{d}H_2, \\ G_3^*\nu_2 &= \nu_3 + \mathrm{d}H_3, \quad G_4^*\nu_3 = \nu_4 + \mathrm{d}H_4. \end{aligned} \tag{22}$$

Lemma 9.1 (Algebra of exact comparisons)**.** *If $F^*\alpha = \beta + \mathrm{d}h$ and $G^*\beta = \gamma + \mathrm{d}k$, then*

$$(F \circ G)^*\alpha = \gamma + \mathrm{d}(k + h \circ G).$$

If $F^*\alpha = \beta + \mathrm{d}h$, then

$$(F^{-1})^*\beta = \alpha + \mathrm{d}(-h \circ F^{-1}).$$

Proof. The first identity follows by pulling the first equality back by G, substituting the second equality, and using $G^*\mathrm{d}h = \mathrm{d}(h \circ G)$. For the second, pull the supplied identity back by F^{-1} and rearrange. These calculations also fix the signs of the gauges under inversion. □

Set

$$P_2 = G_2 \circ G_1, \qquad P_3 = G_3^{-1} \circ P_2, \qquad \Psi = G_4^{-1} \circ P_3.$$

Successive applications of Lemma 9.1 to (22) yield

$$\Psi^*\beta = \lambda + \mathrm{d}f, \qquad f = H_1 + H_2 \circ G_1 - H_3 \circ P_3 - H_4 \circ \Psi.$$

Therefore the witnesses for Theorem 1.1 are

$$F = \Psi^{-1} = G_1^{-1} \circ G_2^{-1} \circ G_3 \circ G_4, \qquad h = -f \circ F. \tag{23}$$

They satisfy $F^*\lambda = \beta + \mathrm{d}h$, and exterior differentiation gives $F^*\omega = \Omega$.

An explicit identity isotopy is available without choosing junctions between separate time intervals:

$$F_t = G_{1,t}^{-1} \circ G_{2,t}^{-1} \circ G_{3,t} \circ G_{4,t}.$$

Every factor and its inverse is jointly smooth. The inverse family is the reverse-order composition of the inverse factors and is jointly smooth as well. Since every $G_{j,0} = \mathrm{id}$, this family starts at the identity and ends at the F in (23). This proves the required smooth isotopy and completes the theorem.

10 Cylindrical compatible structures

We prove the cylindrical assertion in the companion statement. Fix its Liouville domain, completion, boundary Y, and form α. The boundary restriction of the symplectic form is $\mathrm{d}\alpha$. Outward transversality of the Liouville field makes α contact: the restriction of $\mathrm{d}\alpha$ to $\xi = \ker \alpha$ is nondegenerate. The normalized Reeb field is the unique solution of $\alpha(R_\alpha) = 1$ and $\iota_{R_\alpha}\mathrm{d}\alpha = 0$. In local frames this is an invertible linear system with smooth coefficients, so its solutions form a smooth field. The constant-rank kernel ξ is an actual smooth vector subbundle of TY.

10.1 Smooth symplectic normalization

Let Ω be a smooth nondegenerate alternating form on a real vector bundle and let h be a smooth positive metric. Define A by

$$h(Au, v) = \Omega(u, v).$$

Then A is invertible and h-skew-adjoint. The operator $S = -A^2 = A^*A$ is positive and self-adjoint. Put

$$P = S^{1/2}, \qquad J = AP^{-1}.$$

The positive square root commutes with A, hence $J^2 = -\mathrm{id}$. Furthermore,

$$\Omega(u, Jv) = h(Au, AP^{-1}v) = h(Pu, v),$$

which is symmetric and strictly positive for $u = v \neq 0$. Since J is h-orthogonal and commutes with A, it preserves Ω as well.

These constructions are smooth in the base point. To see the only nonpolynomial step, in local smooth orthonormal frames the derivative of the square map at a positive matrix P is $H \mapsto PH + HP$. In an eigenbasis it multiplies the (i, j) entry by $p_i + p_j > 0$, so it is invertible. The implicit function theorem gives local smooth positive square roots; uniqueness

makes them agree on overlaps. Inversion is smooth on invertible matrices. This also gives the intrinsic construction independently of the chosen frames.

There is an exact recovery property. If J_0 is already compatible and $h(u,v) = \Omega(u, J_0 v)$, then $h(J_0 u, v) = \Omega(u,v)$, so $A = J_0$, $P = \mathrm{id}$, and the construction returns J_0. This equality, not just a homotopy class, will preserve the prescribed end.

Applied to ξ with $\Omega = \mathrm{d}\alpha|_\xi$, the construction produces at least one smooth compatible j_ξ: choose a smooth positive metric by gluing local chart metrics with a partition of unity, then normalize. Smoothness of the native restriction and inclusion identifies this bundle construction with the literal contact distribution. Empty bases and zero-rank fibers cause no exception to the construction.

10.2 The prescribed operator on a neighborhood of the end

Now fix any prescribed smooth compatible j_ξ. Split a product tangent vector as

$$(v,s) = bR_\alpha + \eta + s\partial_t, \qquad b = \alpha(v), \quad \eta \in \xi,$$

and set

$$J_{\mathrm{end}}(v,s) = (j_\xi \eta + sR_\alpha, -b).$$

Its square is minus the identity. On product coordinates the actual ambient form pulls back to

$$E^* \mathrm{d}\beta = e^t(\mathrm{d}t \wedge \alpha + \mathrm{d}\alpha).$$

The Reeb annihilation identity and compatibility of j_ξ give

$$E^*\mathrm{d}\beta\big((v,s), J_{\mathrm{end}}(v,s)\big) = e^t\big(s^2 + b^2 + \mathrm{d}\alpha(\eta, j_\xi\eta)\big) > 0$$

for nonzero (v,s). The same decomposition proves preservation of the two-form and symmetry of the induced pairing.

Use the completion's flow and end immersion to transport this operator to its open escaping basin. Uniqueness of the flow and outward transversality give the native basin coordinates; the differential and its inverse identify the actual tangent bundles. Thus the transported operator and $h_{\mathrm{end}} = \mathrm{d}\beta(\,\cdot\,, J_{\mathrm{end}}\,\cdot\,)$ are smooth on an open neighborhood of the entire closed end $K = E(Y \times [0,\infty))$. In particular, this neighborhood extends past the seam; a formula defined only for $t > 0$ would not suffice. The completion geometry proves that K is closed and lies in that open basin.

10.3 Relative extension and recovery on the whole closed end

We use the following relative metric fact: if K is closed, U is open with $K \subset U$, and a smooth positive metric q is given on U, then a global smooth positive metric h exists with $h = q$ on every point of K. Here K need not be compact.

For completeness, use the convex set of positive symmetric forms in each tangent fiber, additionally requiring equality to q over K. Near a point of K, the prescribed q supplies a local section. Near a point outside K, choose a chart neighborhood disjoint from K and use a positive chart metric. A locally finite smooth partition of unity glues these sections. Positivity is preserved by convex combinations, and every contributing section equals q at points of K. This proves the required equality also at the boundary of K. Finite-dimensional native fiber topology gives the bounded unit-ball property required to register the result as a Riemannian metric.

Apply this fact to h_{end} and normalize the resulting global h against $\Omega = \mathrm{d}\beta$. Smooth normalization gives a global compatible J, and the recovery property gives exactly J_{end} on all of K. Define the output metric by $g = \mathrm{d}\beta(\,\cdot\,, J\,\cdot\,)$; it is smooth and positive by the preceding calculation. The auxiliary h is not identified with g away from K.

This proves arbitrary prescribed extension, including $t = 0$. Supplying the contact structure constructed above proves unconditional cylindrical existence. The same construction works on the quotient supplied by the native finite-type adapter: use its domain, completed primitive, and end map throughout. Its source identifications remain unchanged. The comparison application then chooses this J before quantifying the weakly tamed target form and retains the original growth hypothesis.

11 The finite-type Liouville application

Use the supplied domain and smooth completion realization from Section 4 of the companion *Definitions, hypotheses, and target theorems*: the compact-piece inclusion j, complete reference flow φ_t, and actual end immersion E are part of the data. In particular, $E(y,t) = \varphi_t(j(b(y)))$, $E^*\beta = e^t\alpha$, and the embedded nonnegative end together with $j(W)$ covers X, with positive times disjoint from $j(W)$. Write $r = e^t$ on this end. Then $\beta = r\alpha$ and $\partial_t = r\partial_r$. The preceding construction lets us choose a compatible cylindrical J with $J(r\partial_r) = R_\alpha$ and an r-independent compatible structure on $\ker\alpha$.

On the end, decompose a tangent vector as $v = a\,r\partial_r + bR_\alpha + \xi$ with $\xi \in \ker\alpha$. Then $Jv = aR_\alpha - b\,r\partial_r + J\xi$, and hence $-\mathrm{d}r(Jv) = br = \beta(v)$. Moreover,

$$\mathrm{d}\beta = \mathrm{d}r \wedge \alpha + r\,\mathrm{d}\alpha, \qquad g(r\partial_r, r\partial_r) = r.$$

The radial line is orthogonal to the Reeb and contact directions. It follows that $\nabla_g r = r\partial_r$, so

$$\beta = -\mathrm{d}r \circ J, \qquad |\mathrm{d}r|^2_{g^*} = r.$$

Choose a smooth cutoff χ on $[1,\infty)$ equal to zero near 1 and one for $r \geq 2$, with values in $[0,1]$. Define $R = 1 + \chi(r)(r-1)$ on the end and $R = 1$ on the compact piece. It glues smoothly, is at least one, equals r for $r \geq 2$, and is at most two on the remaining compact region. It is proper because Y and W are compact and $R = r$ far out. Thus the radial hypotheses hold with $c = 1$ and $r_* = 2$.

If $\omega = \mathrm{d}\lambda$ weakly tames this J and obeys the inverse-form estimate $CB(r)$ for sufficiently large r, it obeys the main theorem's estimate $CB(R)$ outside a compact set. Theorem 1.1 gives the required exact comparison. The reference completion already carries its prescribed complete all-real-time Liouville flow; on the end its field is $r\partial_r$, whose logarithmic-radius coordinate has unit speed. No completeness of the other primitive is used in this application.

12 The properly embedded Stein application

Let X be a smooth real manifold, let J be a smooth almost complex structure, and let $\iota : X \hookrightarrow \mathbb{C}^N$ be a smooth proper topological embedding with injective differential and $\mathrm{d}\iota \circ J = I \circ \mathrm{d}\iota$. These are the explicit input data of the companion statement; no separate complex atlas or integrability premise is used in the proof. Write the ambient coordinates as $z_\nu = x_\nu + iy_\nu$ and define

$$\beta_{\mathrm{E}} = \frac{1}{2}\sum_{\nu=1}^{N}(x_\nu\,\mathrm{d}y_\nu - y_\nu\,\mathrm{d}x_\nu), \qquad \Omega_{\mathrm{E}} = \mathrm{d}\beta_{\mathrm{E}} = \sum_{\nu=1}^{N}\mathrm{d}x_\nu \wedge \mathrm{d}y_\nu.$$

If I is the standard complex structure, then $\beta_{\mathrm{E}} = -\frac{1}{4}\mathrm{d}\|z\|^2 \circ I$ and $\Omega_{\mathrm{E}}(u, Iv) = \langle u, v\rangle_{\mathbb{R}}$.

Put $R = e + \|\iota\|^2$. The assumed differential identity gives

$$\beta := -\frac{1}{4}\mathrm{d}R \circ J = \iota^*\beta_{\mathrm{E}}, \qquad \Omega = \mathrm{d}\beta = \iota^*\Omega_{\mathrm{E}}. \tag{24}$$

Consequently

$$\Omega(v, Jw) = \Omega_{\mathrm{E}}(\mathrm{d}\iota(v), I\,\mathrm{d}\iota(w)) = \langle \mathrm{d}\iota(v), \mathrm{d}\iota(w)\rangle_{\mathbb{R}}\,.$$

Because ι is an immersion, this is positive definite. It is smooth in x, so it is exactly the smooth induced Euclidean metric g. The identity also proves nondegeneracy of Ω; closedness follows from (24). Compatibility with J follows from the corresponding ambient identities. Thus the reference is determined from the embedding.

Properness of ι implies properness of R: a bounded R-sublevel is the inverse image of a closed Euclidean ball, and hence compact. For every tangent vector,

$$|\mathrm{d}R(v)| = 2|\,\langle \iota(x), \mathrm{d}\iota(v)\rangle_{\mathbb{R}}\,| \le 2\|\iota(x)\|\,|v|_g.$$

Taking the g-dual norm gives

$$|\mathrm{d}R|^2_{g^*} \le 4\|\iota(x)\|^2 = 4(R - e).$$

Therefore the radial identity holds everywhere with $c = 1/4$, and $c|\mathrm{d}R|^2_{g^*} \le R - e < R$. Any fixed $r_* > 1$ may be used for the main theorem's exterior formulation.

It remains to align the lower endpoint of the growth function. Suppose B is admissible on $[e, \infty)$. Choose a smooth step ζ equal to zero for $r \le e + 1$ and one for $r \ge e + 2$. Define

$$\widetilde{B}(r) = \begin{cases} r, & 1 \le r \le e + 1, \\ r + \zeta(r)(B(r) - r), & r > e + 1. \end{cases}$$

Smoothness of the step at its constant region makes this smooth across the join. It is positive and at least r, and equals B for $r \ge e + 2$. Changing the integrand on a bounded interval preserves divergence of $\int r/\widetilde{B}(r)^2\,\mathrm{d}r$. Enlarging the exceptional compact set by $\{R \le e + 2\}$ preserves the inverse-form bound. Theorem 1.1 now applies with $\widetilde{B}$.

This proves the Stein-reference corollary without a finite-type hypothesis. The geometric verification concerns the given embedding and its induced reference; it does not require choosing a new Stein manifold or proving an embedding existence theorem.

13 The common-reference consequence

Suppose λ_1 and λ_2 both satisfy the main theorem for one fixed radial reference. Apply the theorem separately, retaining their independent growth witnesses, to obtain

$$F_i^*\lambda_i = \beta + \mathrm{d}h_i \qquad (i = 1, 2).$$

Pull the first equality back by F_2^{-1} and use the inverse of the second equality. The result is

$$(F_1 \circ F_2^{-1})^*\lambda_1 = \lambda_2 + \mathrm{d}\big((h_1 - h_2) \circ F_2^{-1}\big).$$

Taking exterior derivatives gives the symplectic equality. Composing the identity isotopy of F_1 with the inverse isotopy of F_2 gives the required smooth isotopy of the same comparison map. Thus both primitives are compared with the same reference, while their constants, functions B_i, and compact exceptions remain independent.

14 Where the hypotheses enter

Hypothesis	Role in the proof
Compatibility and explicit g	Give the radial metric formula, dual norms, and the orthogonality of J used in all estimates.
Weak, nonstrict taming	Gives the resolvent contraction for the first path and positivity after adding a compatible radial form for the other paths.
Properness of R	Bounds the primitive on each compact sublevel and converts scalar trajectory bounds into compact confinement.
Radial identities only on the exterior	Give the exterior estimates; compactness supplies the missing bounds on the unrestricted core.
Inverse-form bound and divergent integral	Construct k and the proper first-path scalar $\log S(R)$.
No primitive growth bound	The coefficient A is constructed to dominate the actual primitive, with no upper growth restriction.
Both time directions	Produce actual inverse endpoint maps; reverse interpolation uses the moving barrier Q rather than a forward estimate alone.
Joint smoothness	Gives smooth diffeomorphisms, the Cartan calculation, the integral gauge, and the final identity isotopy.

All four flows are finite-parameter Moser evolutions. The general theorem does not turn their existence into a claim that the autonomous Liouville fields of the prescribed primitives are complete. Its conclusion is precisely the exact primitive identity, its exterior-derivative consequence, and the smooth identity isotopy stated at the beginning.